Getting a Piece Out

Probability during a Ludo opening turn

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The rule being analysed

A player begins with no piece outside the starting area. We use the following precise version of the rule:

1. The player gets up to **three attempts to obtain the first six**. If all three rolls are non-sixes, the turn ends and no piece comes out.
2. On obtaining the first six, the player brings a piece out and receives an extra roll. A second consecutive six gives another extra roll.
3. A non-six after one or two sixes ends the turn, and the piece remains out.
4. **Three consecutive sixes cancel the entire turn, including bringing the piece out.** Thus the piece does not remain out in this case.

These are the rules assumed for this calculation, rather than a claim about every version of Ludo. We assume that the player chooses to bring a piece out when permitted.

What are we finding?

We seek the probability that a piece is out **at the end of the opening turn**. Merely rolling a six is not enough: a later third consecutive six can cancel the result.

Write S for a six and F for any non-six. Put

$$P(S) = p, \qquad P(F) = q = 1 - p.$$

Assume successive rolls are independent, with the same probabilities on every roll. For a fair die, $p = 1/6$ and $q = 5/6$.

The title must say *opening turn*, not *first throw*. The opening turn can contain up to five rolls, as in FFSSF or FFSSS.

Sequences that leave a piece out

There may be zero, one or two initial failures before the first six. After that six, a successful turn must end in either SF or SSF: one or two consecutive sixes followed by a non-six.

Each row below is a complete possible turn. Independence allows us to multiply roll probabilities along a row.

	Complete sequence	Probability
1	SF	pq
2	SSF	p^2q
3	FSF	pq^2
4	FSSF	p^2q^2
5	FFSF	pq^3
6	FFSSF	p^2q^3

These six cases are mutually exclusive, so their probabilities add. If E denotes "a piece remains out at the end of the turn", then

$$\begin{aligned} P(E) &= pq + p^2q + pq^2 + p^2q^2 + pq^3 + p^2q^3 \\ &= pq(1 + q + q^2) + p^2q(1 + q + q^2) \\ &= \boxed{pq(1 + p)(1 + q + q^2)}. \end{aligned}$$

The factor $1 + q + q^2$ accounts for zero, one or two initial non-sixes. The factor $pq + p^2q$ accounts for the two acceptable endings once that initial waiting period is over.

Sequences that leave no piece out

There are two ways to fail: obtain no six in the three starting attempts, or obtain three consecutive sixes after zero, one or two initial failures.

Complete sequence	Probability
FFF	q^3
SSS	p^3
FSSS	p^3q
FFSSS	p^3q^2

Therefore,

$$\boxed{P(E^c) = q^3 + p^3(1 + q + q^2)}.$$

Checking that all possibilities are covered

Either the first three attempts are all non-sixes, or the first six occurs at position 1, 2 or 3. In the latter case, the turn then finishes with SF, SSF or SSS, counted from that first six. These alternatives exhaust the rule and never overlap.

The algebra confirms this. Using $p + q = 1$,

$$P(E) + P(E^c) = (1 + q + q^2)\big[pq(1 + p) + p^3\big] + q^3.$$

Simplify the expression in brackets first:

$$\begin{aligned} pq(1 + p) + p^3 &= pq + p^2q + p^3 \\ &= pq + p^2(q + p) \\ &= pq + p^2 \\ &= p(q + p) = p. \end{aligned}$$

Hence

$$\begin{aligned} P(E) + P(E^c) &= p(1 + q + q^2) + q^3 \\ &= (1 - q)(1 + q + q^2) + q^3 \\ &= 1 - q^3 + q^3 = 1. \end{aligned}$$

For a fair die

Substitute $p = 1/6$ and $q = 5/6$:

$$\begin{aligned} P(E) &= \frac{1}{6} \cdot \frac{5}{6} \cdot \frac{7}{6} \left(1 + \frac{5}{6} + \frac{25}{36}\right) \\ &= \frac{35}{216} \cdot \frac{91}{36} = \boxed{\frac{3185}{7776} \approx 40.96\%}, \end{aligned}$$

$$P(E^c) = 1 - \frac{3185}{7776} = \boxed{\frac{4591}{7776} \approx 59.04\%}.$$

As a second check, the chance of obtaining the first six within three attempts is $1 - q^3$. Of these turns, the proportion that subsequently produces two more sixes is p^2. Thus

$$P(E) = (1 - q^3)(1 - p^2),$$

which agrees with $pq(1 + p)(1 + q + q^2)$ because $1 - q = p$ and $1 - p = q$.

A Python experiment

The **theoretical probability** is the exact probability under our rule and fair-die assumptions. The **experimental probability** is the fraction of simulated turns that leave a piece out:

$$\widehat{P}(E) = \frac{\text{number of successful turns}}{\text{number of simulated turns}}.$$

The program rolls a simulated die and follows the stopping rule. It does not choose success directly using the theoretical answer. Each call to `piece_out` simulates one complete opening turn.

```
import random
from itertools import product


def piece_out(roll):
    # Up to three attempts to obtain the first six.
    for attempt in range(3):
        if roll() == 6:
            break
    else:
        return False

    # One six has occurred. A non-six now ends a valid turn.
    for extra in range(2):
        if roll() != 6:
            return True
    return False  # Three consecutive sixes cancel the turn.


def experiment():
    rng = random.Random(20261030)
    theoretical = 3185 / 7776
    checkpoints = {1000, 10000, 100000, 1000000}
    successes = 0
    print("Turns,Successes,Experimental,Theoretical,Error_pp")
    for n in range(1, 1000001):
        successes += piece_out(lambda: rng.randint(1, 6))
        if n in checkpoints:
            estimate = successes / n
            error = 100 * (estimate - theoretical)
            print(f"{n},{successes},{estimate:.8f},"
                  f"{theoretical:.8f},{error:+.5f}")
```

To run this experiment from the displayed code, add:

```
experiment()
```

The accompanying file also includes the rule tests and runs both automatically. From its folder, run `python3 ludo_experiment.py`. It needs only Python's standard library.

Results from an actual run

The following results were obtained by running the supplied script with seed `20261030`. Rows are cumulative checkpoints from the same run, not four separate experiments. The theoretical success probability is

$$P(E) = \frac{3185}{7776} = 0.4095936214\ldots \approx 40.959362\%.$$

Turns	Successes	Experimental	Error (pp)
1,000	399	39.9000%	−1.05936
10,000	4,050	40.5000%	−0.45936
100,000	40,961	40.9610%	+0.00164
1,000,000	409,796	40.9796%	+0.02024

Here "pp" means *percentage points*. The signed error is

$$100\big(\widehat{P}(E) - P(E)\big).$$

After one million turns, 409,796 were successful and 590,204 were unsuccessful. Thus the experimental failure probability is 59.0204%, compared with the theoretical 59.040638%.

Why the answers are close, but not identical

A finite random sample need not reproduce the exact probability. A different seed generally gives different counts; the fixed seed makes this run reproducible, not a proof.

With many independent turns, the observed fraction tends to settle near the theoretical probability. It need not get closer at every checkpoint: here 100,000 turns happened to give a closer estimate than 1,000,000. More trials improve typical accuracy, not accuracy at every step.

Testing that the program follows the rule

The script checks all six successful and four unsuccessful terminal sequences listed earlier, including the correct stopping point.

It also checks all $6^5 = 7,776$ equally likely five-roll records. Every opening turn needs at most five rolls; unused trailing rolls are ignored. Exactly 3,185 records succeed, confirming $P(E) = 3185/7776$. This exhaustive check verifies the code for our stated rule; the simulation illustrates random variation.